

Form Number : _____

Paper Code
(0000CMD302119006)



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2019 - 2020)

PRE-MEDICAL : ENTHUSIAST COURSE

PHASE - I

Test Type : **COMMON TEST (SRG)**

Test Pattern : **NEET(UG)**

TEST DATE : 18 - 08 - 2019

Time : 3 Hours

Maximum Marks : 720

Important Instructions

Do not open this Test Booklet until you are asked to do so

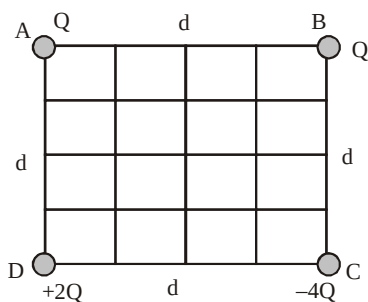
1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Student can not use log tables and calculators or any other material in the examination hall.
3. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
4. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
5. Each correct answer carries 4 marks, while one mark will be deducted for each incorrect answer.
6. Use the OMR provided separately for answering the question.

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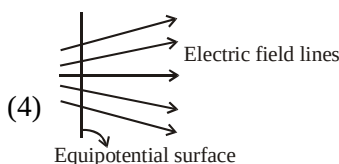
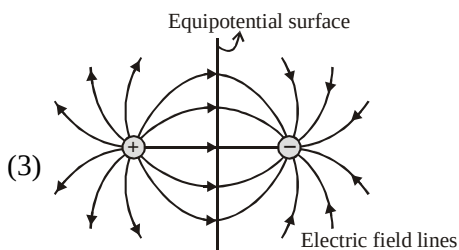
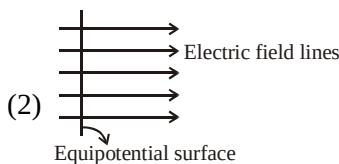
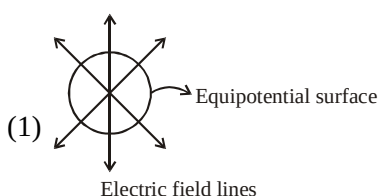
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1. If magnitude of force applied by charge placed at point B on A is F , then choose the incorrect statement :-

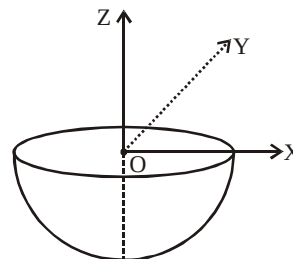


- (1) Magnitude of force applied by B on C is $4F$.
- (2) Magnitude of force applied by B on D is F .
- (3) Magnitude of force applied by A on C is $2F$.
- (4) Magnitude of force applied by B on D is $\frac{F}{\sqrt{2}}$.

2. If we move in the direction perpendicular to the electric field, work done by the electric field will be zero and surface thus generated is called equipotential surface. Choose the incorrect matching :-



3. A uniformly charged hemispherical shell has radius R . The curvature centre is taken as origin. The total charge on shell is Q . If field at point A $(0, 0, -Z_0)$ is $\vec{E}$, then field at point B $(0, 0, Z_0)$ is $(Z_0 < R)$:-

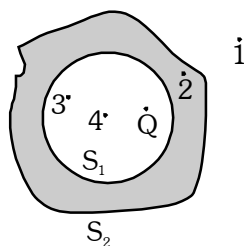


- (1) $\vec{E}$
- (2) $-\vec{E}$
- (3) $\vec{E} - \frac{kQ}{z_0} \hat{k}$
- (4) $\vec{E} + \frac{kQ}{z_0} \hat{k}$

4. Suppose you have a charged particle A and a charged particle B separated by $4m$. The charge on each particle is 8 nC although the sign of the charge on particle A is positive and that on particle B is negative. The charge of particle A is then increased by a factor of 4, the charge of particle B decreased by a factor of 0.5 and the separation between the two particles decreased by a factor of 0.5. The Coulomb force on each of the particles is approximately.

- (1) $6 \times 10^{-7} \text{ N}$
- (2) $1 \times 10^{-7} \text{ N}$
- (3) $3 \times 10^{-7} \text{ N}$
- (4) $4 \times 10^{-7} \text{ N}$

5. For an isolated neutral conductor having spherical cavity and charge Q inside the cavity as shown in figure. Choose the correct option from column-II to column-I.



	Column-I		Column-II
(P)	Net electric field due to all the charges inside the cavity is non-zero at the point	1	Point-1
(Q)	Electric field due to induced charge on surface S_1 is zero at the point	2	Point -2
(R)	Electric field due to induced charge on surface S_2 is non zero at the point	3	Point -4
(S)	Electric field due to charge Q is zero at the point	4	none of above

Codes :

P	Q	R	S
(1) 1	2	3	4
(2) 3	4	1	4
(3) 3	3	1	4
(4) 1	4	1	4

6. In which of the following case(s), flux defined normally through the area given will be non-zero.

($\vec{E}$ represents electric field while $\vec{A}$ represents area vector)

- (1) $\vec{E} = 3\hat{i} + 4\hat{j}$, $\vec{A} = 5\hat{k}$
- (2) $\vec{E} = 3\hat{i}$, $\vec{A} = 4\hat{j}$
- (3) $\vec{E} = 2\hat{i} + \hat{j}$, $\vec{A} = -2\hat{i} + 4\hat{j}$
- (4) $\vec{E} = 2\hat{k}$, $\vec{A} = 5\hat{k}$

7. Consider a region where charge density varies with radius as:- (here ρ_0 is a constant)

$$\rho = \rho_0 r \quad 0 \leq r \leq R$$

$$= 0 \quad r > R$$

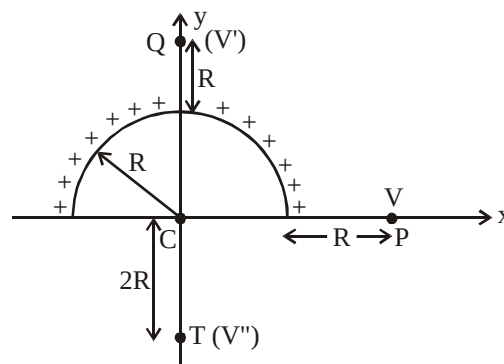
Choose the correct statement :-

- (1) Electric field varies as r^2 for $r < R$
- (2) Electric field is zero for $r > R$
- (3) Electric field varies as r for $r < R$
- (4) Electric field varies as $\frac{1}{r}$ for $r > R$

8. Point charges $-2q$, $-2q$ and $+q$ are put on the vertices of an equilateral triangle of side a . Find the work done by some external force in increasing the separation to $2a$

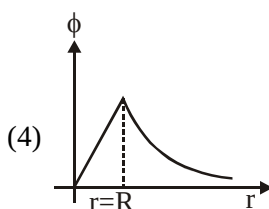
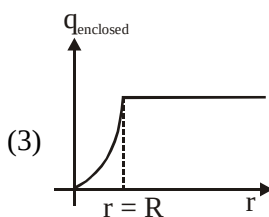
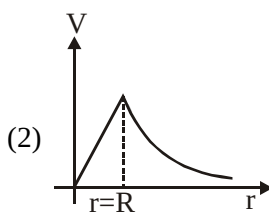
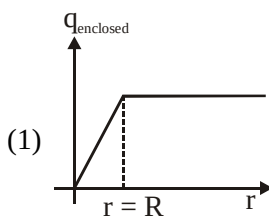
- (1) 0
- (2) $\frac{4kq^2}{a}$
- (3) $-\frac{4kq^2}{a}$
- (4) $\frac{kq^2}{2a}$

9. Charge is uniformly distributed on the curved surface of hemisphere. Potential at P, Q and T is V , V' and V'' respectively, then select correct option



- (1) $V'' < V < V'$
- (2) $V' < V < V''$
- (3) $V' = V = V''$
- (4) $V' = V < V''$

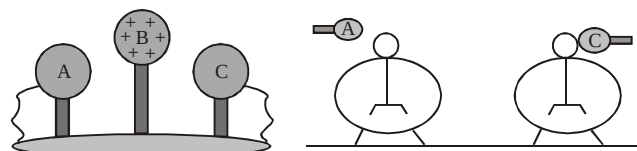
10. Graph shows magnitude of electric potential (V), charge enclosed (q_{enclosed}) within the concentric sphere and net flux (ϕ) through a concentric spherical Gaussian surface as a function of distance (r) from centre of a uniformly positive charged solid sphere of radius R . Choose the **CORRECT** option(s):-



11. Which of the following not represent an electrostatic field :-

- (1) $\vec{E} = y\hat{i} + x\hat{j}$
- (2) $\vec{E} = y\hat{i}$
- (3) $\vec{E} = y\hat{i} + x\hat{j} + z\hat{k}$
- (4) $\vec{E} = \alpha\vec{r}$ (α is constant)

12. Positively charged sphere B is placed between two neutral spheres A and C. We cut connection of A and C with ground. If we put A closer to the first electroscope and touch C to the sphere of second electroscope, then choose the **CORRECT** statement(s)



- (1) Charge on electroscope leaves kept near A will be negative
- (2) Charge on electroscope leaves kept near A will be positive
- (3) No charge on electroscope leaves kept in contact with C.
- (4) Charge on electroscope leaves kept in contact with C will be positive.

13. Column-I gives the variation of electric intensity at P described in column II as a function of r . Column-II specifies a point at distance r from the centre/axis of a symmetrical distribution of charge. Match the columns.

	Column-I		Column-II
(A)	$E \propto \frac{1}{r^2}$	(P)	P lies outside a cylinder having uniform volume charge density.
(B)	$E \propto \frac{1}{r}$	(Q)	P lies inside a spherical charged conductor.
(C)	$E \propto r$	(R)	P lies inside a spherical body having uniform volume charge density.
(D)	$E \propto r^0$	(S)	P lies outside a spherical body having uniform volume charge density.

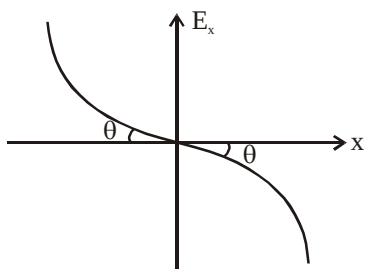
Codes :

A	B	C	D
(1) S	P	R	Q
(2) P	S	R	Q
(3) R	Q	P	S
(4) S	R	P	Q

14. A material have a charge density of $6 \mu\text{C}/\text{m}^3$. The material is in the form of sphere and is kept in an external field $0.01 \text{ N/C } \hat{i}$. Then choose the correct statement.

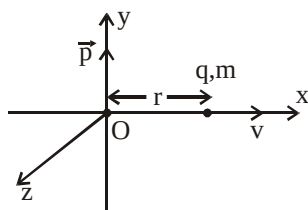
- (1) The field lines will be perpendicular to the surface of sphere.
- (2) The charge redistribution will take place till field inside the sphere is zero.
- (3) The field will not be perpendicular to the surface.
- (4) The field lines can't be present inside the sphere.

15. Variation of x-component of electric field with x-coordinate in a region is given in the graph. A small electric dipole having dipole moment $P_0 \hat{i}$ is placed at origin, then net force on the dipole is:



- (1) $P_0 \tan \theta$ along positive x-axis
- (2) Zero
- (3) $P_0 \tan \theta$ along negative x-axis
- (4) $P_0 \tan \theta$ along positive y-axis

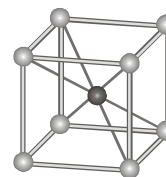
16. An electric dipole is fixed along y-axis with its centre at origin. At a given instant a charge particle is moving along x-axis then



- (1) Power of electric force on charge particle is zero.
- (2) Acceleration and velocity are parallel
- (3) Time rate of change in velocity is zero.
- (4) If charge is free to move, it always moves along x-axis.

Paragraph for Questions no. 17 and 18

Cesium Chloride is a salt with a crystal structure in which cubes of Cs^+ ions (side length $a \sim 0.4 \text{ nm}$) surround Cl^- ions, as pictured below. This structure is called BCC structure.



17. What is the magnitude of the net electrostatic force on the Cl^- ions due to its eight nearest neighbor Cs^+ ions?

- (1) $1.9 \times 10^{-9} \text{ N}$
- (2) $1.9\sqrt{2} \times 10^{-9} \text{ N}$
- (3) 0
- (4) $1.9\sqrt{3} \times 10^{-9} \text{ N}$

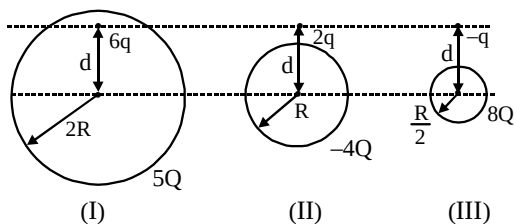
18. Occasionally defects arise in which one of the Cs^+ ions is missing. We call this a vacancy. In this case, what is the magnitude and direction (relative to the vacancy) of the net electrostatic force on the Cl^- ion due to its remaining seven?

- (1) $1.9 \times 10^{-9} \text{ N}$
- (2) $1.9\sqrt{2} \times 10^{-9} \text{ N}$
- (3) 0
- (4) $3.8 \times 10^{-9} \text{ N}$

19. A charge q is placed at a distance of $2R$ from the centre of an uncharged spherical conducting shell of radius R . Find the radius of neutral ring on the shell

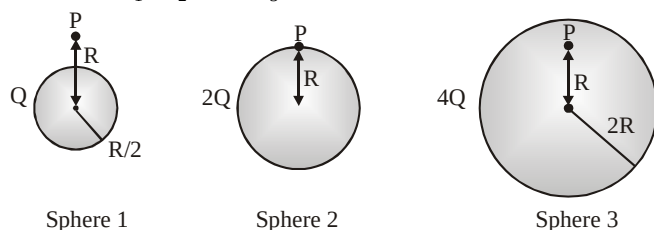
- | | |
|---------------------------|--------------------------|
| (1) R | (2) $\frac{R}{2}$ |
| (3) $\frac{\sqrt{3}R}{2}$ | (4) $\frac{R}{\sqrt{2}}$ |

20. Figure, shown above, shows three situations involving a charged particle and a uniformly charged spherical shell. The charges and radii of the shells are indicated in the figure. If F_1 , F_2 and F_3 are the magnitudes of the forces on the particle due to the presence of the shell in situations (I), (II) and (III) respectively then



- (1) $F_3 > F_2 > F_1$ (2) $F_2 > F_2 = F_3$
 (3) $F_3 = F_2 > F_1$ (4) $F_1 > F_2 > F_3$

21. Charges Q , $2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$ respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then



- (1) $E_1 > E_2 > E_3$ (2) $E_3 > E_1 > E_2$
 (3) $E_2 > E_1 > E_3$ (4) $E_3 > E_2 > E_1$

22. Let $E_1(r)$, $E_2(r)$ and $E_3(r)$ be the respective electric fields at a distance r from a point charge Q , an infinitely long wire with constant linear charge density λ , and an infinite plane with uniform surface charge density σ . If $E_1(r_0) = E_2(r_0) = E_3(r_0)$ at a given distance r_0 , then :-

- (1) $Q = 4\sigma\pi r_0^2$
 (2) $r_0 = \frac{\lambda}{2\pi\sigma}$
 (3) $E_1(r_0/2) = 2E_2(r_0/2)$
 (4) $E_2(r_0/2) = 4E_3(r_0/2)$

23. A uniform electric field exists in xy plane. The potential of points $A(2, 2)$, $B(-2, 2)$ and $C(2, 4)$ are $4V$, $16V$, $12V$ respectively. The electric field is :-

- (1) $(4\hat{i} + 5\hat{j})V/m$ (2) $(3\hat{i} + 4\hat{j})V/m$
 (3) $-(3\hat{i} + 4\hat{j})V/m$ (4) $(3\hat{i} - 4\hat{j})V/m$

24. A particle of mass m and charge Q is placed in an electric field E which varies with time t as $E = E_0 \sin \omega t$.

- (1) The particle executes simple harmonic motion about a position which backward in direction of electric field with uniform velocity.
 (2) The particle executes simple harmonic motion about a position which advances in direction of electric field with uniform acceleration.
 (3) Maximum speed attained by the charge

particle is $\frac{2QE_0}{m\omega}$

- (4) Amplitude of the simple harmonic motion is

$\frac{QE_0}{m^2\omega^2}$

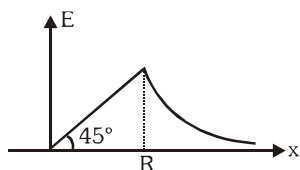
25. A dipole of dipole moment $\vec{P} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at point $A(2, -3, 1)$. The electric potential due to this dipole at the point $B(4, -1, 0)$ is $(ab) \times 10^9$ volt here 'a' represents sign (for negative answer select 0 for positive answer select 1 as first digit of your answer) and 'b' is a single digit that will appear in your answer (For Example : If your answer is -2×10^9 volt then your answer should be 02). All the parameters specified here are in S.I. units.

- (1) 12 (2) 02 (3) 03 (4) 06

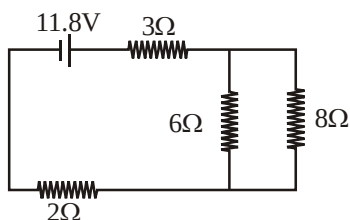
26. Two point charges exert a force F_0 on each other when placed in vacuum. Now the charges are increased to four times, separation between them is doubled and the system is placed in an insulating medium. Now they experience the same force. What should be the dielectric constant of the medium?

- (1) 1 (2) 2 (3) 3 (4) 4

27. There is a solid non conducting uniformly charged sphere having radius $(18)^{1/3} \times 10^3$ m. In figure radial distance v/s electric field graph is given then find the total charge on the sphere?

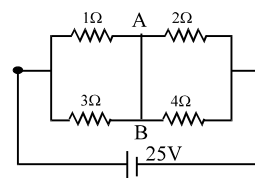


- (1) 3 C (2) 2 C (3) 1 C (4) 4 C
28. There are 8 drops of a conducting fluid. Each has radius r and they are charged to potential 1 volt. They are then combined to form a bigger drop. Find potential of big drop.
- (1) 1 V (2) 2 V (3) 3 V (4) 4 V
29. An electric field $(-3\hat{i} + 2\hat{j}) \text{ Vm}^{-1}$ exists in the space. If potential at the origin is zero then find the potential at (5m,3m).
- (1) 9 V (2) 8 V (3) 7 V (4) 6 V
30. A conducting sphere is placed in air of dielectric strength 3×10^6 volt/m. The minimum radius of the sphere that can be raised to a potential of 9 million volts is :-
- (1) 1 m (2) 2 m (3) 3 m (4) 4 m
31. In the network, the current in the resistor 6Ω is

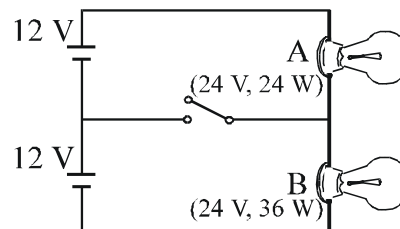


- (1) $\frac{7}{5}$ A (2) $\frac{4}{5}$ A
- (3) $\frac{3}{5}$ A (4) zero

32. Find the current flowing through the segment AB of the circuit shown in figure.



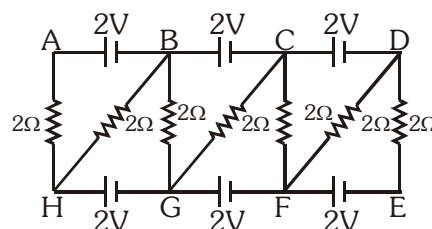
- (1) 1A from A to B (2) 2A from A to B
(3) 1A from B to A (4) 2A from B to A
33. Two light bulbs shown in the circuit have ratings A (24 V, 24 W) and B (24 V and 36 W) as shown. When the switch is closed.



- (1) The intensity of light bulb A increases
(2) The intensity of light bulb A decreases
(3) The intensity of light bulb B remain unchange
(4) The intensity of light bulb B decreases

Paragraph for Question 34 to 36

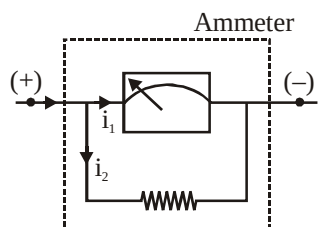
In given circuit, 7 resistors of resistance 2Ω each and 6 batteries of 2V each, are joined together.



34. The potential difference $V_D - V_E$ is—
- (1) $\frac{5}{6}$ V (2) $\frac{6}{7}$ V (3) $-\frac{5}{6}$ V (4) $-\frac{22}{9}$ V
35. The current through branch BG is—
- (1) 1A (2) 0.2A (3) 0.6 A (4) 0.4A
36. The current through battery between A & B is—
- (1) 0.4 A (2) 0.8 A (3) 0.6 A (4) 1 A

Paragraph for Question no 37 and 38

Ammeter is a device to measure an electric current. In order to convert a galvanometer into an ammeter, a resistance having small value is connected in parallel with the galvanometer coil. This resistor is called the shunt. The current to be measured is passed through the ammeter by connecting it in series with the segment which carries the current. When the ammeter is connected in a segment of circuit, the resistance of the segment increases, which reduces the main current to be measured. To minimize the error, the equivalent resistance of ammeter circuit should be small. This is one reason why the shunt having a small resistance r is connected in parallel to the coil. A galvanometer of resistance $100\ \Omega$ gives a full scale deflection for a current of 1 mA .



37. Shunt required to convert it into an ammeter giving full scale deflection for a current of 10 amp is

(1) $\frac{1}{100}\ \Omega$ (2) $\frac{100}{9999}\ \Omega$ (3) $\frac{100}{1999}\ \Omega$ (4) $\frac{100}{10001}\ \Omega$

38. When the current through ammeter is giving full scale deflection then heat generated by shunt per unit time is

- (1) 1 W (2) 0.9999 W
(3) 1.0001 W (4) no heat will be dissipated

Paragraph for question nos. 39 to 41

A small insect crawls in the direction of electron drift along bare copper wire that carries a current of 2.56 A . It travels with the drift speed of the electron in the wire of uniform cross section area 1 mm^2 . Number of free electrons for copper $= 8 \times 10^{22}/\text{cc}$ & resistivity of copper $= 1.6 \times 10^{-8}\ \Omega\text{m}$.

39. How much time would the insect take to crawl 1.0 cm if it crawls at the drift speed of the electrons in the wire?

- (1) 50 sec (2) 5 sec
(3) 5000 sec (4) None of these

40. What is order of the average time of collision for free electrons of copper?

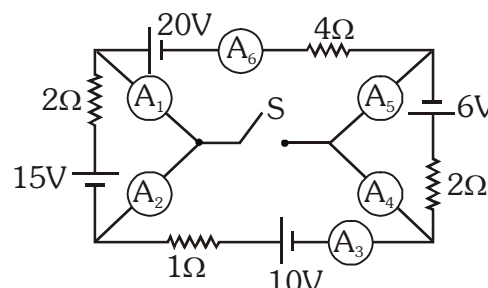
- (1) 10^{-14} sec (2) 10^{-17} sec
(3) 10^{-11} sec (4) 10^{-8} sec

41. If the insect starts from the point of zero potential at $t=0$, it reaches a point of ____ potential after 10 sec .

- (1) $80\ \mu\text{V}$ (2) $-80\ \mu\text{V}$
(3) $160\ \mu\text{V}$ (4) $-160\ \mu\text{V}$

Paragraph for Question 42 to 44

In circuit shown all ammeters are ideal



42. If switch S is open, the ammeter (s) that read more than 10 A

- (1) A_1 & A_2 (2) A_3 & A_4
(3) A_5 & A_6 (4) None of these

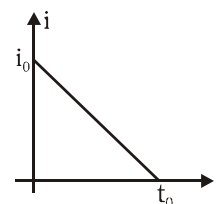
43. If switch S is closed, the ammeters that read less than 3 A

- (1) A_1 (2) A_2 (3) A_3 (4) A_4

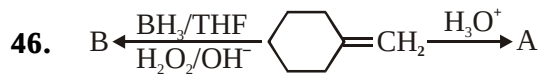
44. If switch S is closed. The ammeters that show decrease in reading

- (1) A_1 (2) A_2 (3) A_5 (4) A_6

45. Relation between current i in conductor and time t is shown in figure. If the resistance of conductor is R , then total heat dissipated across resistance R is :-



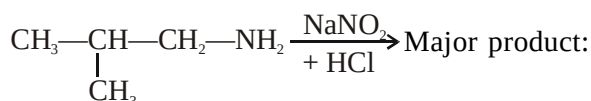
- (1) $\frac{i_0^2 R t_0}{2}$ (2) $\frac{i_0^2 R t_0}{4}$
(3) $\frac{i_0^2 R t_0}{3}$ (4) $i_0^2 R t_0$



A and B are :-

- (1) Both
- (2) Both
- (3)
- (4)

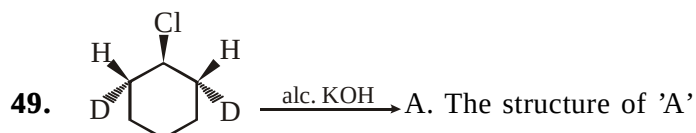
47. Identify major product



- (1) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{OH}$ (2) CH_3-OH
- (3) $\text{CH}_3-\text{C}(\text{OH})(\text{CH}_3)-\text{CH}_3$ (4) $\text{CH}_3-\text{CH}_2-\text{OH}$

48. Order of nucleophilicity in polar protic solvent

- (1) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
- (2) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$
- (3) $\text{I}^- > \text{Cl}^- > \text{Br}^- > \text{F}^-$
- (4) $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$

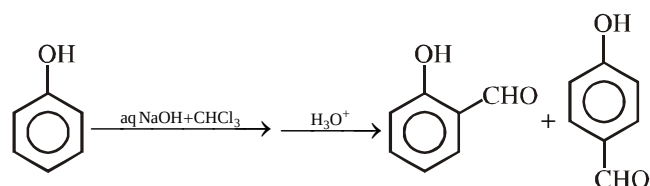


- (1)
- (2)
- (3)
- (4)

50. Compare rate of reaction with Ag^+NO_3^- or rate of $\text{S}_{\text{N}}1$ reaction :-

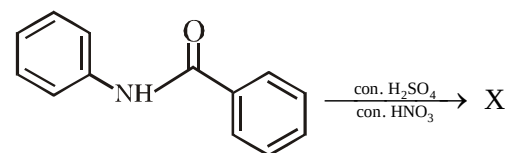
- (a) (b) (c)
- (1) $a > c > b$ (2) $b > c > a$
- (3) $a > b > c$ (4) $c > a > b$

51. Which of the following statement is incorrect in relation of given reaction ?



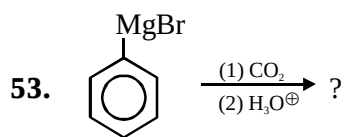
- (1) :CCl_2 is the electrophile in this reaction
- (2) Reaction is an example of electrophilic aromatic substitution reaction
- (3) is formed as an intermediate of this reaction
- (4) $^+\text{CHCl}_2$ is the electrophile of this reaction

52. In the following reaction,

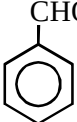
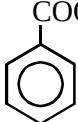
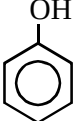


the structure of the major product 'X' is :-

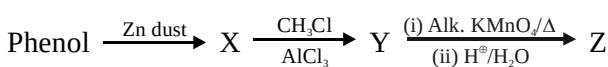
- (1)
- (2)
- (3)
- (4)



Product in above reaction is :-

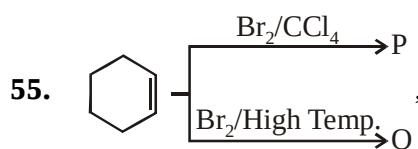
- (1)  (2) 
 (3)  (4) 

54. Consider the following reaction :

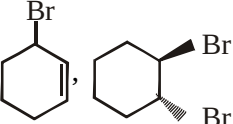
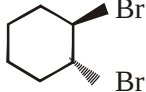
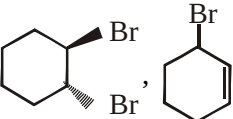
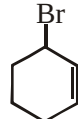


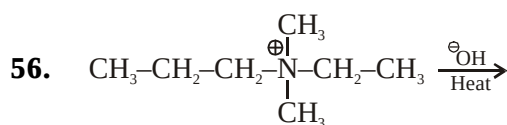
Product 'Z' is :-

- (1) Benzene (2) Toluene
 (3) Benzaldehyde (4) Benzoic acid



P and Q are respectively :-

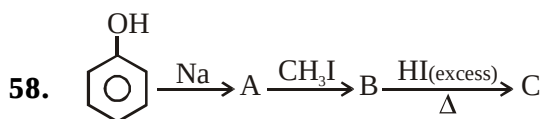
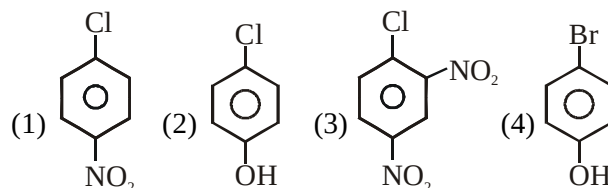
- (1)  (2) P = Q = 
 (3)  (4) P = Q = 



Major product will be ?

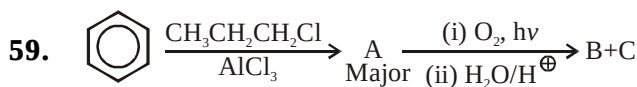
- (1) $\text{CH}_3\text{--CH=CH}_2$ (2) $\text{CH}_3\text{--CH}_2\text{--CH}_3$
 (3) $\text{CH}_2=\text{CH}_2$ (4) $\text{CH}_3\text{--CH}_3$

57. Which of the following is most reactive towards nucleophilic substitution with aq.KOH



Final product of the reaction is/are :-

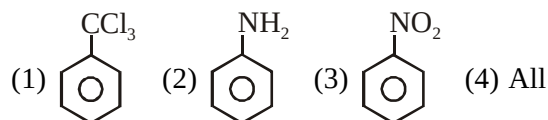
- (1) Only  (2)  + CH_3I
 (3)  + CH_3I (4)  + CH_3OH



Products B & C are :-

- (1) $\text{CH}_3\text{CH}_2\text{OH}$ & $\text{CH}_3\text{--}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{--CH}_3$
 (2) Ph--OH & $\text{CH}_3\text{--}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{--CH}_3$
 (3) $\text{Ph--}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{--CH}_3$ & CH_3OH
 (4) Ph--COOH & $\text{CH}_3\text{--}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{--CH}_3$

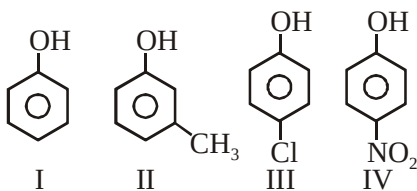
60. Which of the following will not give Friedel Crafts reaction ?



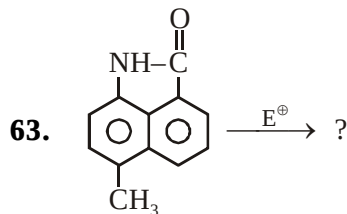
61. How many structural monochloroderivaties are possible for iso-pentane ?

- (1) 2 (2) 3 (3) 4 (4) 5

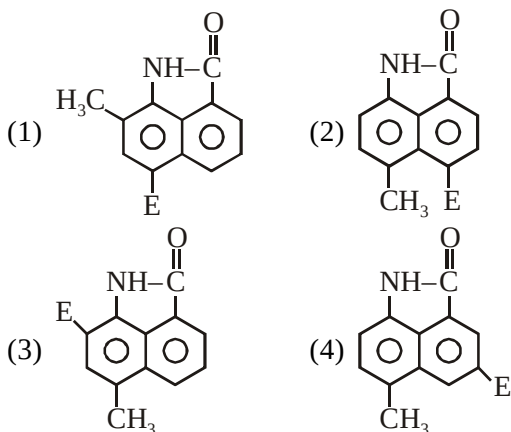
62. Correct order of reactivity of ESR is :-



- (1) III > I > II > IV (2) II > III > IV > I
(3) II > IV > I > III (4) II > I > III > IV



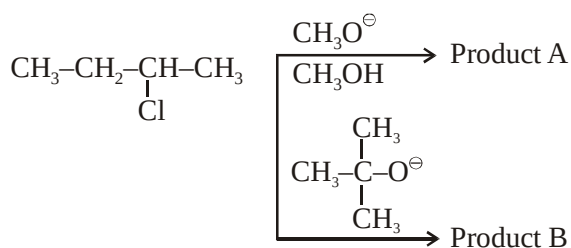
Identify the product :-



64. In nitration of benzene, con. H_2SO_4 act as :-

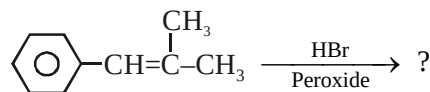
- (1) Base (2) Catalyst
(3) Reducing Agent (4) Acid

65. Identify relationship between product A & B

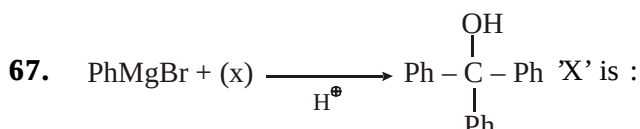


- (1) Chain isomer
(2) Position isomer
(3) Functional isomer
(4) Not an isomer

66. Identify the product

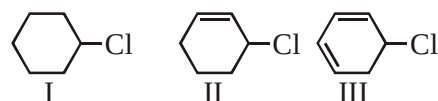


- (1) (2)
(3) (4)

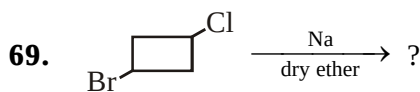


- (1) $\text{Ph}-\overset{\text{O}}{\parallel}{C}-\text{H}$ (2) $\text{Ph}-\overset{\text{O}}{\parallel}{C}-\text{Ph}$
(3) $\text{Ph}-\underset{\text{Ph}}{\underset{|}{CH}}-\overset{\text{O}}{\parallel}{C}-\text{H}$ (4) $\text{Ph}-\overset{\text{O}}{\parallel}{C}-\text{CH}_3$

68. Reactivity of dehydrohalogenation is :-

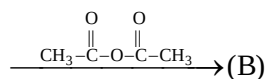
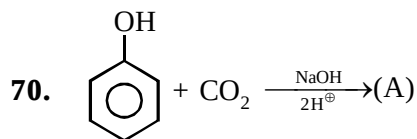


- (1) I > II > III (2) I > III > II
(3) III > II > I (4) III > I > II

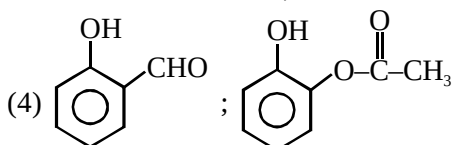
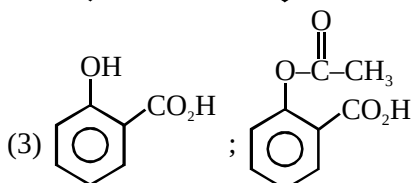
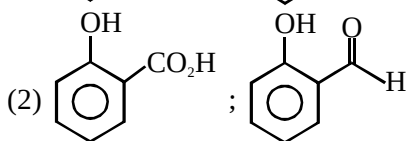
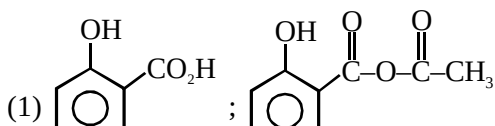


Identify the product ?

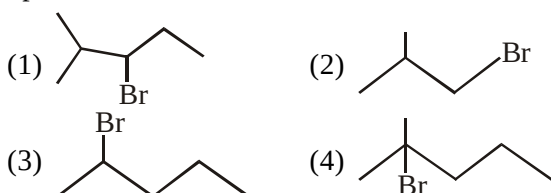
- (1) (2)
(3) (4)



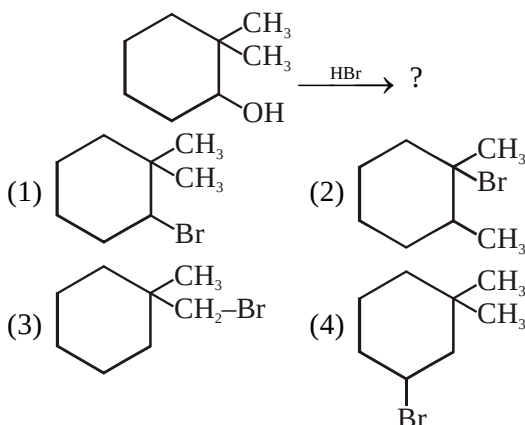
Product (A) and (B) respectively are :



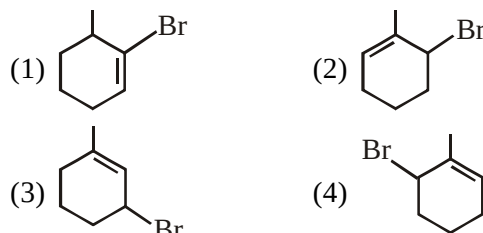
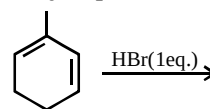
71. Which alkyl bromide will yield only one alkene upon E² elimination ?



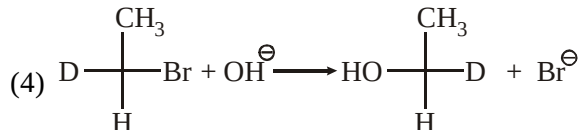
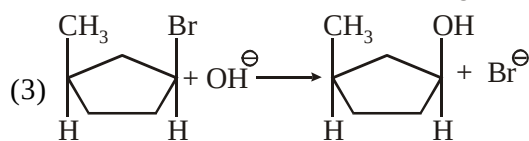
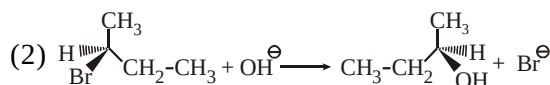
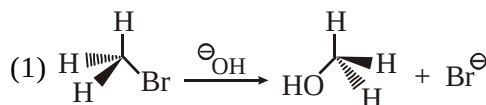
72. What is the major product of the following reaction ?



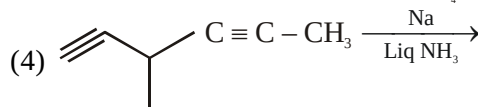
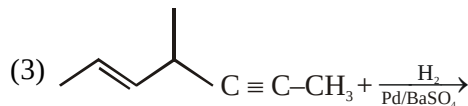
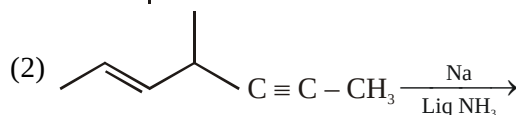
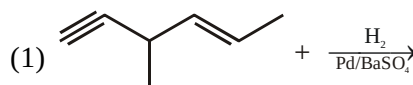
73. Major product of following reaction is :-



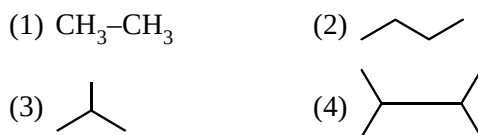
74. Which of the following is not S_N2 reaction ?



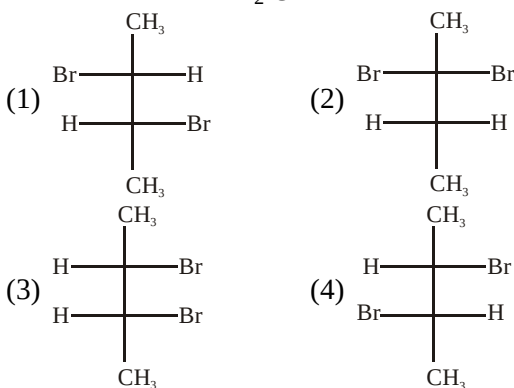
75. Which of the following compounds will lose optical activity after the reaction?



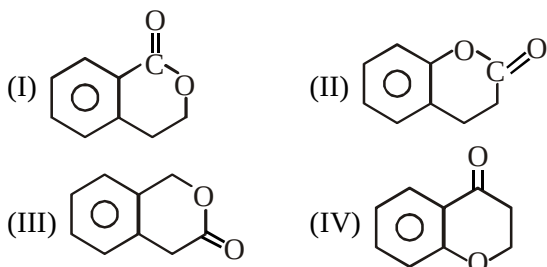
76. Which of the following can not be prepared by wurtz reaction :-



77. Trans-2-butene + Br₂ gives -



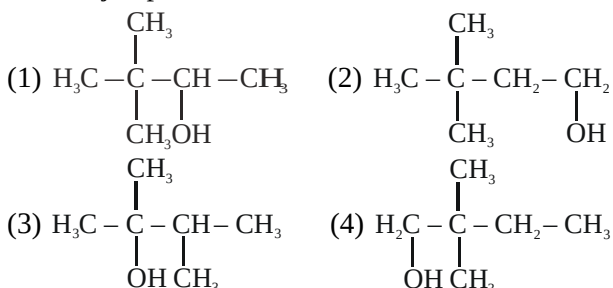
78. Correct order of electrophilic substitution reaction is:-



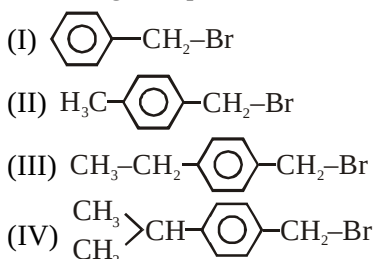
- (1) IV > I > II > III (2) II > IV > III > I
(3) III > II > I > IV (4) IV > II > I > III

79. $\text{H}_3\text{C}-\text{C}(\text{CH}_3)_2-\text{CH}=\text{CH}_2 \xrightarrow{\text{H}_2\text{O}/\text{H}^+}$ major product

The major product is :-

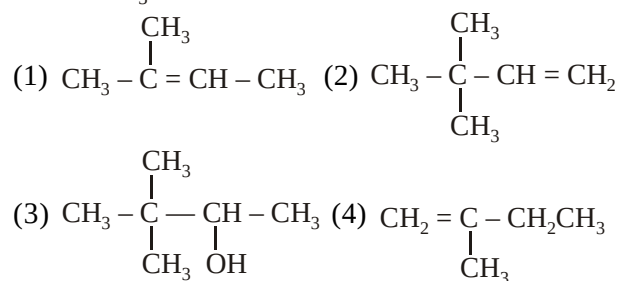


80. Arrange in decreasing order towards hydrolysis of following compounds with SN¹ reactions :-

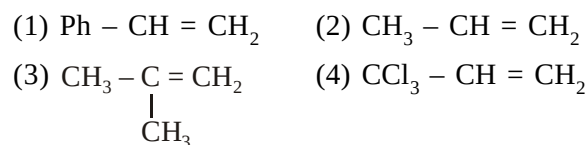


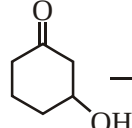
- (1) III > IV > II > I (2) IV > III > II > I
(3) II > III > IV > I (4) I > II > III > IV

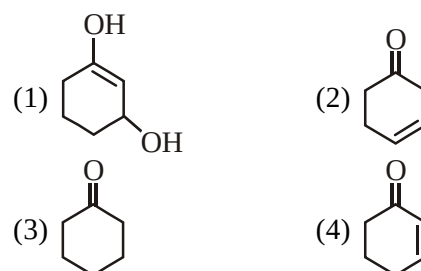
81. $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{CH}(\text{Cl})-\text{CH}_3 \xrightarrow[\Delta]{\text{Alc. KOH}}$?



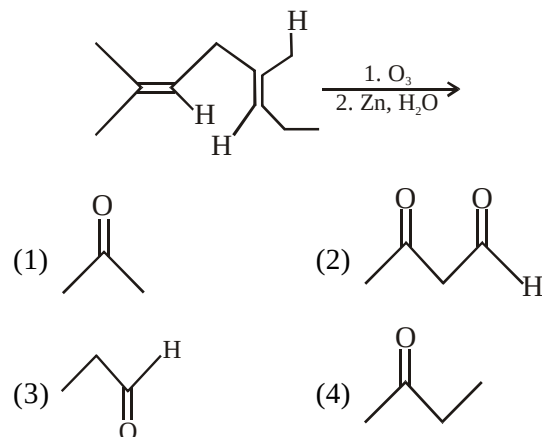
82. In which of the following compound addition of HCl does not take place according to markovnikov rule.



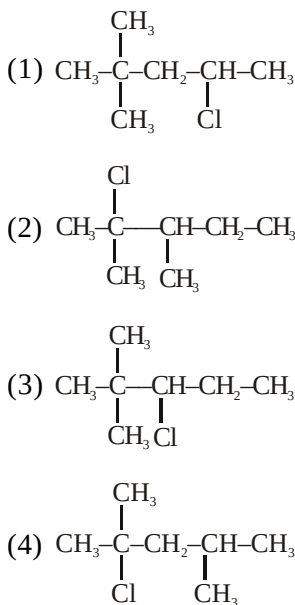
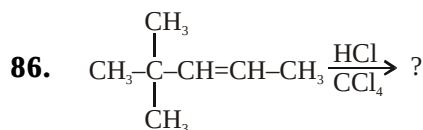
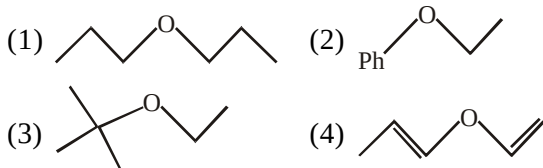
83.  $\xrightarrow[\Delta]{\text{H}^+}$ (Major product)



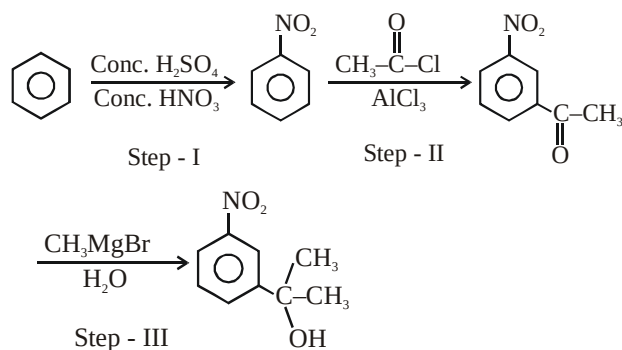
84. Which product will not form during this reaction?



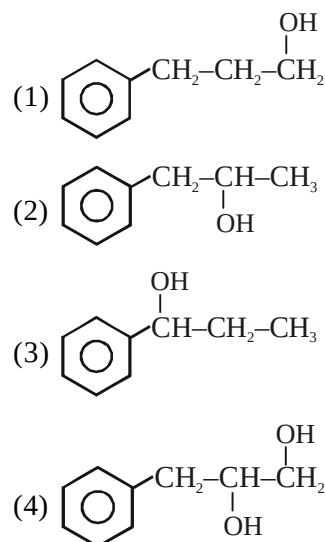
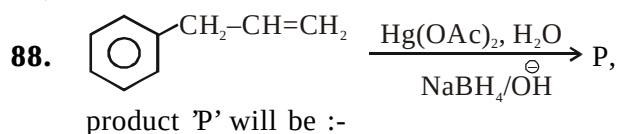
85. Which ether can not be prepared by williamson synthesis :-



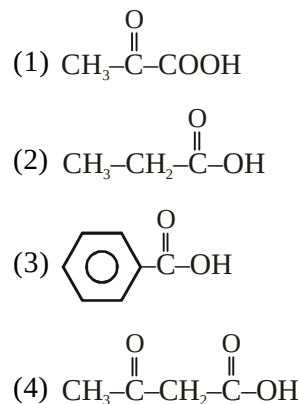
87. In the following reaction which of the following step is wrong :-



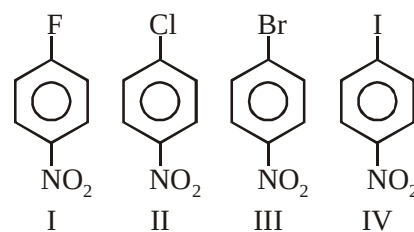
- (1) Step - I
 (2) Step - II
 (3) Step - III
 (4) Step - I and II both



89. Which of the following carboxylic acid undergoes decarboxylation most easily on heating :-



90. Compare rate of reaction of following compound with NaOH :-



- (1) IV > III > II > I
 (2) I > II > III > IV
 (3) I = II = III = IV
 (4) IV > I > II > III

91. Which of the following character of garden pea is present on 4th chromosome.

- (1) Plant height, pod colour, cotyledon colour
- (2) Plant height, flower position, pod shape
- (3) Pod colour, seed colour, flower colour
- (4) Pod shape, seed shape, plant height

92. In a cross between $TtRr \times TtRr$, total 640 seeds are produced. Find the number of plants produced with one dominant and other recessive traits :-

- (1) 106
- (2) 300
- (3) 240
- (4) 360

93. The ratio of childrens with blood groups O : AB : B : A born to a set of parents in which mother is with B blood group and father with A blood group, will be :-

- (1) 1 : 1 : 1 : 1
- (2) 2 : 0 : 2 : 0
- (3) 0 : 0 : 4 : 0
- (4) All of these

94. Grain colour in wheat is determined by two pairs of polygenes. Following cross $AABB$ (dark colour) $\times$ $aabb$ (light colour), in F_2 generation what proportion of the progeny is likely to resemble either parents :-

- (1) 20%
- (2) 50%
- (3) <15%
- (4) 37.5%

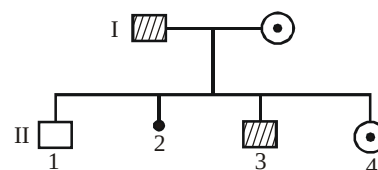
95. In a cross between individuals homozygous for (a and b) and wild type. In this cross 800 out of 1000 individuals were of parental type. Then the distance between a and b :-

- (1) 70 map unit
- (2) 35 map unit
- (3) 20 map unit
- (4) 15 map unit

96. In the population of 50,000 individuals, if the frequency of recessive allele for a character is 40%, then find the number of organisms having dominant character :-

- (1) 30,000
- (2) 4200
- (3) 42,000
- (4) 36,000

97. What is genotype of II-2 if gene is represented by h.



- (1) $x^h x$
- (2) $x^h x^h$
- (3) $x^h y$
- (4) xx

98. In a given plant long leaves S and green veins Y, are dominant respectively over short leaves ss and yellow veins yy. The cross $SSYY \times ssyy$ produce F_1 $SsYy$ when F_1 plant were inbred. The F_2 consisted of 570 long green and 190 short yellow. It show :

- (1) Independent assortment
- (2) Incomplete linkage
- (3) Complete linkage
- (4) Monohybrid cross

99. A hypothetical series of 20 multiple allele is known for certain locus. How many genotypic classes are possible ?

- (1) Genotypic classes = 210
- (2) Genotypic classes = 20
- (3) Genotypic classes = 10
- (4) Genotypic classes = 420

100. Find out phenotypic categories in following cross:

- (A) $AaBbCC \times aaBBcc$
- (B) $GgRrTt \times ggrrTT$
- (C) $aabbCC \times AABBcc$

	A	B	C
(1)	2	4	1
(2)	2	2	2
(3)	2	3	1
(4)	3	2	2

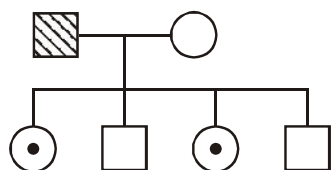
101. Gene P and Q are necessary for normal hearing A deaf boy marries with deaf girl and all their children had normal hearing. The genotype of the parent are :

- (1) $ppqq \times ppQq$
- (2) $ppQQ \times ppQq$
- (3) $ppqq \times PPqq$
- (4) $PPqq \times ppQQ$

102. If two persons with 'AB' blood group marry and have sufficiently large number of children, these children could be classified as 'A' blood group : 'AB' blood group 'B' blood group in 1 : 2 : 1 ratio. Modern technique of protein electrophoresis reveals presence of both 'A' and 'B' type proteins in 'AB' blood group individuals. This is an example of :

- (1) Complete dominance
- (2) Codominance
- (3) Incomplete dominance
- (4) Partial dominance

103. Given pedigree chart shows which type of inheritance.



- (1) Autosomal recessive (Albinism)
- (2) X-linked recessive (colour blindness)
- (3) Y-linked
- (4) Autosomal dominant

104. Which suspect would you charge with the crime?

Victim	Crime scene sample	Suspect-1	Suspect-2
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

- (1) Both suspect 1 & 2
- (2) Only suspect 1
- (3) Only suspect 2
- (4) Neither suspect 1 nor suspect 2

105. Repressible operon system is usually found in (i) pathways. The pathway's end product serves as a (ii) to activate the repressor, turn off enzyme synthesis and prevent overproduction of the end product of the pathway. Genes for this operon are usually switched (iii) and the repressor is synthesized in an (iv) form.

	(i)	(ii)	(iii)	(iv)
(1)	Anabolic	Corepressor	On	Inactive
(2)	Anabolic	Inducer	Off	Active
(3)	Catabolic	Inducer	Off	Active
(4)	Catabolic	Corepressor	On	Inactive

106. During sickle cell anaemia sickle shaped R.B.C. are develop due to which change :

- (1) Glutamic acid → Valine
- (2) Valine → Glutamic acid
- (3) Proline → Valine
- (4) Valine → Proline

107. According to EFB, "The integration of natural science and organism, cells, parts there of and molecular analogues for products and services is known as :-

- (1) Biochemistry
- (2) Bioinformatics
- (3) Biotechnology
- (4) Biology

108. p^{BR322} which is frequently used as vector for cloning gene is :-

- (1) An original bacterial plasmid
- (2) A viral genome
- (3) A modified bacterial plasmid
- (4) A transposon

109. Select the wrong statement:

- (1) Replication of DNA initiates at promoter site
- (2) First artificial recombinant DNA molecule was constructed in 1972
- (3) Developing a DNA vaccine is an example of biotechnology
- (4) Replication of plasmid is independent of the control of chromosomal DNA in bacteria

110. Which of the following enzyme is used to join DNA fragments ?

- (1) Terminase
- (2) Endonuclease
- (3) Ligase
- (4) DNA polymerase

- 111.** First restriction endonuclease to be discovered was:
 (1) Hind II
 (2) Eco RI
 (3) Bam II
 (4) pBR 322

- 112.** Match the columns and choose correct option:-

Cross		Types of gametes		Types of Phenotype		Types of genotype			Total zygotic combinations
A	Monohybrid	a	4	i	4	p	9	x	64
B	Dihybrid	b	8	ii	2	q	3	y	4
C	Trihybrid	c	2	iii	8	r	27	z	16

- (1) A-a-i-p-x, B-b-ii-q-y, C-c-iii-r-z
 (2) A-c-ii-p-y, B-a-iii-r-x, C-b-i-q-z
 (3) A-b-i-q-y, B-c-ii-p-z, C-a-iii-r-x
 (4) A-c-ii-q-y, B-a-i-p-z, C-b-iii-r-x

- 113.** First artificial DNA molecule was constructed by Cohen and1..... in2..... They used3.... enzyme to isolate the antibiotic resistance gene. They synthesized recombinant DNA by linking the antibiotic resistance gene with4.... DNA of5.... bacteria. Recombinant DNA with newly introduced antibiotic resistance gene was then transferred into another bacteria6....

- (1) 1-Tatum, 3-Restriction, 5-E.coli
 (2) 4-Plasmid, 6-Salmonella
 (3) 3-DNA ligase, 4-Plasmid, 5-E.coli
 (4) 2-1972, 3-Restriction, 6-E.coli

- 114.** In a plant height is controlled by two polygenes A and B. Tallest plant is 56 cm tall and height of shortest plant is 30 cm. What will be the height of plant having genotype AAbb?

- (1) 43 cm (2) 48.5 cm
 (3) 37.5 cm (4) 51 cm

- 115.** A man is heterozygous for Albinism as well as Sickle cell anaemia. What portion of his gametes will contain genes for both diseases ?

- (1) $\frac{1}{4}$ (2) $\frac{2}{4}$ (3) $\frac{1}{3}$ (4) $\frac{1}{8}$

- 116.** Which one is **wrong** statement about genetic material?

- (1) It should be able to express itself in the form of Mendelian characters
 (2) DNA chemically is less reactive and structurally more stable than RNA
 (3) RNA has code for the synthesis of proteins
 (4) Only DNA not RNA are able to mutate

- 117.** If the gene frequency for PTC nontasters is 0.4, then what will be the no. of heterozygotes tasters in a population of 4000 ?

- (1) 1440 (2) 1920
 (3) 1080 (4) 2520

- 118.** $\phi \times 174$ bacteriophage contains :-

- (1) ss DNA (2) ds DNA
 (3) ds RNA (4) ss RNA

- 119.** The initial clue about genetic/chromosomal mechanism of sex determination can be traced back to some of the experiments carried out on Find correct option for the blank.

- (1) Birds (2) Fishes
 (3) Insects (4) Plants

- 120.** The first human hormone produced by recombinant DNA technology is :-

- (1) Insulin (2) Estrogen
 (3) Thyroxin (4) Progesterone

- 121.** When a mixture of different types of restriction endonuclease enzymes are used to act on DNA, then :

- (1) Same kind of non-sticky ends are produced
 (2) Different kinds of sticky ends are produced
 (3) Different kinds of Non-sticky ends are produced
 (4) Same kind of sticky ends are produced

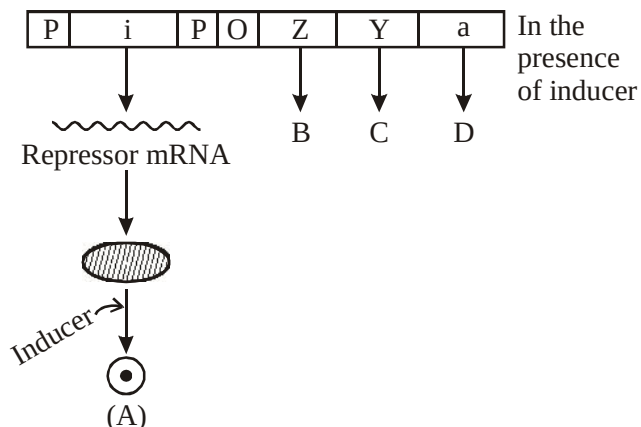
- 122.** The opened up palindromic DNA sequence are joined with their complementarily cut counter parts by:

- (1) Restriction endonuclease enzyme
 (2) DNA polymerase enzyme
 (3) Exonuclease enzyme
 (4) DNA ligase enzyme

123. Out of the following, which is not related to biotechnology?

- (1) Synthesis of a gene
- (2) Correction of a gene
- (3) Transfer of a gene from parent to offspring
- (4) Alternation of a gene

124. In the following diagram A, B, C and D is respectively.



- (1) A - Inactive repressor, B - β galactosidase, C - Permease, D - Transacetylase
- (2) A - Repressor, B - β galactosidase, C - Permease, D - β galactoside
- (3) A - Repressor, B - β galactosidase, C - Transacetylase, D - Permease
- (4) All of these

125. Agarose gel is :

- (1) A natural polymer obtained from protists
- (2) A natural polymer obtained from fungi
- (3) A natural polymer obtained from seaweed
- (4) An artificial polymer

126. Read the following 4 statement (a-d) :-

- (a) Nucleosome contain 200 base pair of DNA
- (b) Nucleosome = Binding DNA + Histone Octamer + Linker DNA + H_1 histone
- (c) 6 Nucleosome units united to form Solenoid structure
- (d) Histone octamer contain $(H_2A H_2B H_3H_1)_2$

How many above statement are **correct** ?

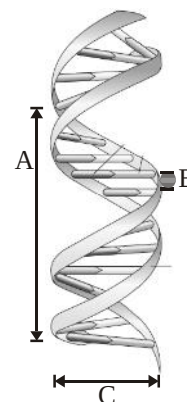
- (1) 4
- (2) 3
- (3) 2
- (4) 1

127. Match column-I with column-II and select the correct option from the codes given below.

Column-I (Codons)		Column-II (Translated amino acid)	
(A)	UUU	(i)	Serine
(B)	GGG	(ii)	Methionine
(C)	UCU	(iii)	Phenylalanine
(D)	CCC	(iv)	Glycine
(E)	AUG	(v)	Proline

- (1) A-(iii), B-(iv), C-(i), D-(v), E-(ii)
- (2) A-(iii), B-(i), C-(iv), D-(v), E-(ii)
- (3) A-(iii), B-(iv), C-(v), D-(i), E-(ii)
- (4) A-(ii), B-(iv), C-(i), D-(v), E-(iii)

128. Given figure represents the DNA double helix model as proposed by Watson and Crick (1953). Select the option that shows **correct** measurements of A, B and C.



- (1) A-3.4 nm, B - 0.34 nm, C - 2 nm
- (2) A-34 nm, B-3.4 nm, C - 20 nm
- (3) A - 3.4 Å, B - 0.34 Å, C - 20 Å
- (4) A - 34 Å, B - 3.4 Å, C - 2 Å

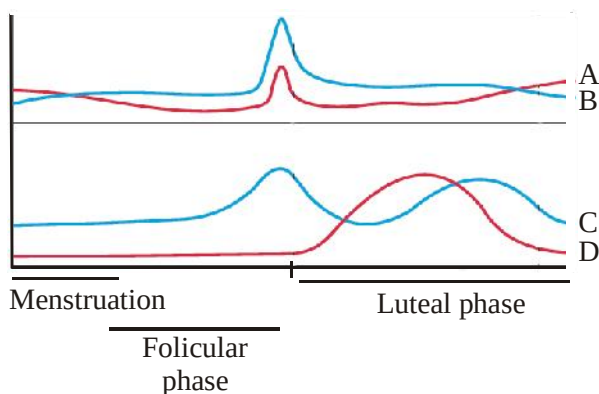
129. Which of the following is Nucleoside ?

- (1) Adenylic Acid
- (2) Adenosine, Uridine
- (3) Adenine, Uracil
- (4) Deoxy, Uridine

- 130.** 5 Methyl uracil is present in which Nucleic Acid
 (1) RNA and DNA
 (2) Only DNA
 (3) Only RNA
 (4) Only double helical RNA
- 131.** Which of the following is sex limited character
 (1) Baldness
 (2) Mammary gland
 (3) Beard Moustache
 (4) Both 2 and 3
- 132.** Choose correct statement from Griffith experiment
 (1) S strain→inject in to mice→Mice died
 (2) R strain→inject in to mice→mice live
 (3) S strain (Heat killed)→Inject in to mice→Mice die
 (4) Both 1 and 2
- 133.** Restriction endonucleases are used in genetic engineering because :-
 (1) They can degrade harmful proteins
 (2) They can join DNA fragments
 (3) They can cut DNA at variable site
 (4) They can cut DNA at specific base sequence
- 134.** A genetically manipulated organism containing in its genome one or more inserted gene of another species is called :-
 (1) Transposon
 (2) Gene expression
 (3) Transgenic
 (4) Retrotransposon
- 135.** Select the incorrect statement :
 (1) Plasmid can replicate inside as well as outside the bacterial cell
 (2) In EcoRI, Co comes from species
 (3) Today we know about more than 900 restriction enzymes in over 230 strains of bacteria
 (4) Agarose gel is obtained from seaweeds

- 136.** How many of the given statements are correct ?
 A. Androgens are produced by sertoli cells
 B. Sperms get nutrition from sertoli cells
 C. Leydig cells are present in ovary
 D. Leydig cells synthesize androgens
 E. Oogenesis takes place in corpus luteum
 F. Menstrual cycle ceases during pregnancy
 G. After one month of pregnancy the embryo's heart is formed.
 (1) Three
 (2) Four
 (3) Six
 (4) Eight
- 137.** Here are some statements about recent amendment in MTP act, find the incorrect one :
 (1) Its intention was to reduce illegal abortions, and consequent maternal mortality and morbidity.
 (2) It was amended in 2017
 (3) If pregnancy has lasted more than 24 weeks but less than 36 weeks, two registered medical practitioners must be of opinion that ground required for MTP exists
 (4) MTP can be performed if there is risk for pregnant woman or the child if born will suffer from physical or mental abnormality as to be seriously handicapped.

138. In the given graph A and B are pituitary hormones while C and D are ovarian hormones. Select the option, which correctly identifies them :



	A	B	C	D
(1)	FSH	LH	Estrogen	Progesterone
(2)	LH	FSH	Estrogen	Progesterone
(3)	FSH	LH	Progesterone	Estrogen
(4)	LH	FSH	Progesterone	Estrogen

139. Which of the following is not a reason for population explosion ?
- Increased health facility along with better living conditions.
 - Increased maternal mortality rate.
 - Rapid decline in death rate
 - Decline in infant mortality rate.
140. Column-I has some contraceptives and Column-II has their ingredients or mode of action. Select the correct match :

	Column-I		Column-II
A	Multiload 375	i	Centchroman
B	Saheli	ii	Progestogens
C	Condom	iii	Sterilization
D	Emergency contraceptive	iv	Copper
E	Tubectomy	v	Latex sheath

- A-iv, B-v, C-i, D-iii, E-ii
- A-iv, B-i, C-v, D-ii, E-iii
- A-iii, B-iv, C-ii, D-i, E-v
- A-ii, B-v, C-iii, D-iv, E-i

141. From the given list, select the hormones whose level is increased during pregnancy but are not secreted by placenta :

FSH, LH, hCG, hPL, estrogen, progesterone, cortisol, prolactin, thyroxine.

- Three
- Four
- Five
- Six

142. Here are some statements about layers of uterus and their roles:

- Endometrium breaks at time of menstruation
- Myometrium has oxytocin receptors and responsible for parturition
- Implantation occur in myometrium
- Myometrium take part in placenta formation.

Select the correct statements :

- A and C
- A and B
- A, B and D
- B, C and D

143. ICSH acts on :

- Sertoli cells, located in seminiferous tubules
- Sertoli cells, located out of seminiferous tubules
- Leydig cells, located in seminiferous tubules
- Leydig cells, located out of seminiferous tubules

144. First polar body is formed during oogenesis at:

- Completion of meiosis I prior to puberty
- Completion of meiosis I after puberty
- Completion of meiosis II prior to puberty
- Completion of meiosis II after puberty

145. Most appropriate assisted reproductive technology for a women, who can not produce ovum but can provide suitable environment for fertilization and further development is :-

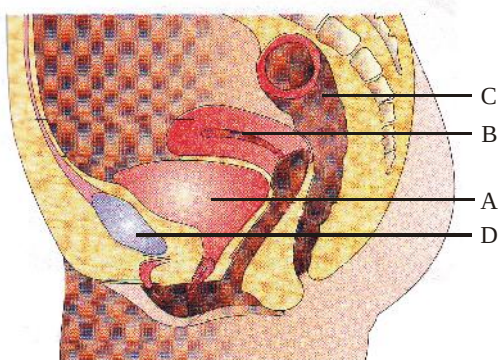
- GIFT
- ZIFT
- ICSI
- IUI

- 146.** India was among the first countries to initiate programmes to attain reproductive health as a social goal. These programmes called ____A____ were initiated in ____B____.

Select the option, which correctly fills A and B.

- (1) A-Reproductive and Child health care, B-1951
(2) A-Reproductive and Child health care, B-1973
(3) A-Family planning, B-1951
(4) A-Family planning, B-1973
- 147.** hCG is similar to which pituitary hormone in its mode of action ?
- (1) FSH (2) LH
(3) Estrogen (4) Progesterone
- 148.** In penis, which of the following structure is covered by fore skin ?
- (1) Glans penis (2) Urethra
(3) Prostate (4) Epididymis

149.



Which of the following labelling represent uterus?

- (1) A (2) B (3) C (4) D
- 150.** Which of the following structures of female external genitalia are covered by public hairs ?
- (1) Labia majora and labia minora
(2) Hymen and clitoris
(3) Mons pubis and labia majora
(4) Labia majora and clitoris

- 151.** How many sperms are ejaculated by human male during one coitus ?

- (1) 20-30 million
(2) 200-300 million
(3) 20-30 lac
(4) 200-300 billion

- 152.** Based on uterine events, follicular phase of menstrual cycle is also known as :-

- (1) Proliferative phase
(2) Luteal phase
(3) Secretory phase
(4) Menstrual phase

- 153.** Site of insemination is :-

- (1) Vagina
(2) Cervix
(3) Uterus
(4) Ampulla

- 154.** By the end of first trimester of embryonic development, which of the following event would have not occurred in embryo ?

- (1) Formation of heart
(2) Formation of major organ systems
(3) Development of limbs and digits
(4) Formation of eye lashes

- 155.** Which of the following pituitary hormone play important role in parturition ?

- (1) Prolactin
(2) Oxytocin
(3) ADH
(4) Relaxin

- 156.** Which of the following is not an intrauterine device (IUD) used for birth control ?

- (1) Multiload 375 (2) LNG-20
(3) Progestasert (4) Implants

- 157.** Which of the following is not an assisted reproductive technology (ART) to help infertile couples ?
- (1) Artificial insemination
 - (2) Zygote intra fallopian transfer
 - (3) Amniocentesis
 - (4) Intra cytoplasmic sperm injection
- 158.** Select incorrect statement about medical termination of pregnancy :-
- (1) Number of MTPs performed in a year all over world is 1/5 of total number of conceivings in a year.
 - (2) MTP is legal in all countries.
 - (3) MTP was legalised in 1971 in India.
 - (4) MTP is to get rid of unwanted pregnancy.
- 159.** Which of the following barrier method of contraception provide protection against STDs ?
- (1) Condom
 - (2) Diaphragms
 - (3) Cervical cap
 - (4) Vault
- 160.** Marriageable age of female is __A__ and that of male is __B__ in India.
- (1) A-18 years, B-20 years
 - (2) A-21 years, B-18 years
 - (3) A-18 years, B-21 years
 - (4) A-20 years, B-18 years
- 161.** Unequal cytokinesis is seen in human :-
- (1) Spermatogenesis, Oogenesis
 - (2) Oogenesis, Cleavage
 - (3) Spermatogenesis, Cleavage
 - (4) Spermatogenesis, Organogenesis

- 162.** Number of chromosomes in spermatid and first polar body in human gonads is :-
- (1) Spermatid - 46, first polar body - 46
 - (2) Spermatid - 23, first polar body - 23
 - (3) Spermatid - 46, first polar body - 23
 - (4) Spermatid - 23, first polar body - 46
- 163.** Capacitation of sperms occur in __A__ and decapacitation occur in __B__ :-
- (1) A - Epididymis, B - Vagina
 - (2) A - Vagina, B - Epididymis
 - (3) A - Vas deferens, B - Epididymis
 - (4) A - Seminiferous tubule , B - Epididymis
- 164.** In which of the following method of contraception, insemination is prevented ?
- (1) Lactational amoenorhoea
 - (2) Premature ejaculation
 - (3) Coitus interruptus
 - (4) Erectile dysfunction
- 165.** Pick up the **correct** match

(A)	I.U.T.	(1)	Embryo transfer done inside uterus after in vitro fertilization
(B)	GIFT	(2)	Taking semen from donor and transferred into female genital tract
(C)	AI	(3)	One sperm transferred into ovum
(D)	ZIFT	(4)	Collect gamete and transfer in female fallopian tube
(E)	I.C.S.I.	(5)	Zygote put into fallopian tube

	A	B	C	D	E
(1)	1	2	5	3	4
(2)	1	3	5	4	2
(3)	1	4	3	5	2
(4)	1	4	2	5	3

166. Read the following four statements (A - D) :-

- (A) Removal of gonads cannot be considered as a contraceptive option
- (B) Complete lactation could help as a natural method of contraception
- (C) Surgical methods of contraception prevent gamete formation
- (D) Medical termination of pregnancy is legalised in our country

How many statements are correct?

- (1) Four (2) Three (3) Two (4) One

167. Pituitary hormone acting on ovary is __A__, acting on uterus is __B__ and ovarian hormone acting on uterus is __C__.

A

B

C

- | | | |
|--------------|----------|----------|
| (1) Oxytocin | FSH | Androgen |
| (2) FSH | Oxytocin | Estrogen |
| (3) LH | Estrogen | Oxytocin |
| (4) FSH | LH | Estrogen |

168. Here are some events that occur just before ovulation in female, select incorrect one :-

- (1) Maturation of ovarian follicle to form one Graafian follicle
- (2) Sudden rise in LH levels due to positive feedback
- (3) Sudden rise in progesterone levels
- (4) Estrogen at its highest level as compared to rest of menstrual cycle

169. Time gap between two set of daily contraceptive pill is :-

- (1) 3 days
- (2) 7 days
- (3) 1 month
- (4) 15 days

170. Which of the following event is not related to embryonic development in second trimester ?

- (1) First movements of the foetus
- (2) Covering of body by fine hairs
- (3) Beginning of heart function
- (4) Growth of foetus

171. Fertilization can only occur if :-

- (1) Ovum and sperms are transported simultaneously to ampulla
- (2) Ovum and sperms are transported simultaneously to epididymis
- (3) Ovum and sperms are transported simultaneously to cervix
- (4) Ovum and sperms are transported simultaneously to ovary

172. Presence of fluid filled cavity antrum is character of :-

- (1) Primary follicle
- (2) Secondary follicle
- (3) Tertiary follicle
- (4) Primordial follicle

173. Select the incorrect statement :-

- (1) MTPs are relatively safe during first trimester
- (2) Contraceptives are not required for normal reproductive health
- (3) After amniocentesis, intracytoplasmic sperm injection is used
- (4) Amniocentesis is banned for sex determination

174. Ultimate aim of assisted reproductive technology (ART) is :-

- (1) Protection from sexually transmitted infections
- (2) Sex determination
- (3) Contraception
- (4) Help infertile couples to have children

175. An ideal contraceptive should be :-

- (i) User friendly (ii) Easily available
- (iii) Effective (iv) Irreversible
- (v) With no or least side effect

- (1) i, iii, iv, v
- (2) i, ii, iii, iv, v
- (3) i, ii, iii, v
- (4) i, ii, iv, v

176. Which one of the following statements about morula in humans is *correct* ?

- (1) It has more cytoplasm and more DNA than an uncleaved zygote
- (2) It has almost equal quantity of cytoplasm as an uncleaved zygote but much more DNA
- (3) It has far *less* cytoplasm as well as *less* DNA than in an uncleaved zygote
- (4) It has more or less equal quantity of cytoplasm and DNA as in uncleaved zygote

177. Identify the **correct** statement on 'inhibin' :-

- (1) Inhibits the secretion of LH, FSH and Prolactin.
- (2) Is produced by granulosa cells in ovary and inhibits the secretion of FSH.
- (3) Is produced by granulosa cells in ovary and inhibits the secretion of LH.
- (4) Is produced by nurse cells in testes and inhibits the secretion of LH.

178. Identical twins will be produced when :

- (1) One spermatozoa fertilizes two ova
- (2) One ovum is fertilized by two spermatozoa
- (3) Two eggs are fertilized
- (4) One fertilized egg divides into two blastomeres and they become separate

179. Which of the following approaches does **not** give the defined action of contraceptive ?

(1)	Barrier methods	prevent fertilization
(2)	Intra uterine devices	Increase phagocytosis of sperms, suppress sperm motility and fertilizing capacity of sperms
(3)	Hormonal contraceptives	Prevent/retard entry of sperms, prevent ovulation and fertilization
(4)	Vasectomy	Prevents spermatogenesis

180. Given below are four methods (A–D) and their modes of action (a–d) in achieving contraception. Select their correct matching from the four options that follow

Method	Mode of Action
A. The pill	(a) Prevents sperms reaching cervix
B. Condom	(b) Prevents fertilization
C. Vasectomy	(c) Prevents ovulation
D. Copper T	(d) Semen contains no sperms

Matching :-

- (1) A – (c), B – (d), C – (a), D – (b)
- (2) A – (b), B – (c), C – (a), D – (d)
- (3) A – (c), B – (a), C – (d), D – (b)
- (4) A – (d), B – (a), C – (b), D – (c)

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